

Circuit Simulation Lab:25

Frequency Dividers

Introduction

The purpose of this experiment is to introduce the design of Divide by 2 Circuit

D-type Flip-Flop is as a binary divider, for Frequency Division or as a “divide-by-2” counter. Here the inverted output terminal Q (NOT-Q) is connected directly back to the Data input terminal D giving the device “feedback” as shown below.

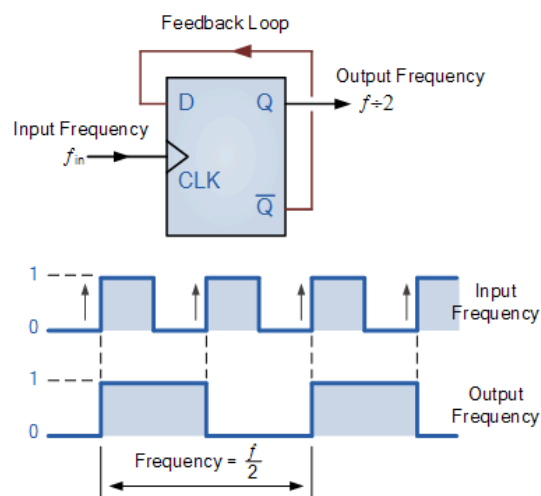
Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado



Divide-by-2 Counter



Design the above Circuit in Verilog HDL and write its testbench and capture the Waveforms,

Use Behavior modeling or Structural Modeling